

Figure 1: General closed loop configuration

1 General solution of a non-ideal closed loop op-amp circuit

Let's solve the circuit in figure 1.
Defining $A(s)$ as the open loop transfer function:

$$A(s) = \frac{A_0}{1 + \frac{s}{\omega_0}} \tag{1}$$

Using Kirchhoff current low:

$$\epsilon = \frac{\frac{V_{in}}{Z_1} + \frac{V_{out}}{Z_2}}{\frac{1}{Z_1} + \frac{1}{Z_2}}$$

and

$$A(s)\epsilon = -V_{out}$$

Therefore

$$V_{out} = -A(s)\frac{\frac{V_{in}}{Z_1} + \frac{V_{out}}{Z_2}}{\frac{1}{Z_1} + \frac{1}{Z_2}}$$

With a bit of algebra, this leads to

$$\frac{V_{out}}{V_{in}} = \frac{-A(s)Z_2}{Z_1 + Z_2 + A(s)Z_1} \tag{2}$$

Finally, expliciting $A(s)$:

$$\frac{V_{out}}{V_{in}} = \frac{-A_0 Z_2}{A_0 Z_1 + (Z_1 + Z_2)\left(1 + \frac{s}{\omega_0}\right)} \tag{3}$$

2 CITIROC circuit solution in Laplace space

2.1 HG preamp

Cutoff pulsation and gain of the HG preamp in open loop mode:

$$\omega_0(HG) = 2\pi \cdot 450\ kHz \tag{4}$$

$$A_0(HG) = 65\ dB = 1778 \tag{5}$$

Open loop transfer function for a non ideal opamp in inverting mode:

$$HG\ ol: \quad TF(s) = \frac{-A_0\omega_0}{s + \omega_0} \tag{6}$$

Same opamp in closed loop mode. Loop closed with two capacitance as in fig. 2. The feedback

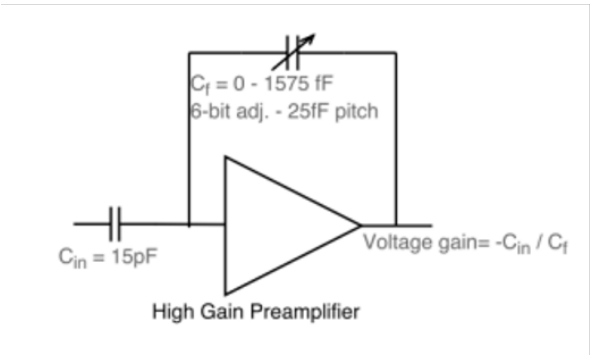


Figure 2: High gain closed loop circuit.

capacitance C_F is determined by slow control parameter and follows

$$C_F = 25\ fF \cdot (63 - DAC)$$

The value of the HG DAC considered is 44, as in sFGD configuration. This corresponds to $C_F = 25\ fF \cdot (63 - 44) = 475\ fF$. This corresponds to:

$$\beta = Z_2/Z_1 = C_{in}/C_F = 31.5 \tag{7}$$

The transfer function of the HG preamp in closed loop (inverted) configuration is:

$$HG\ cl: \quad TF(s) = \frac{-A_0\beta}{1+\beta+A_0+(1+\beta)\frac{s}{\omega_0}} \tag{8}$$

This inverted low pas filter circuit is equivalent to a **first order low-pass filter** with different cutoff frequency and gain, its transfer function can be written as

$$HG\ cl: \quad TF(s) = -G_{HG}\frac{1}{1+\frac{s}{\omega_c}} \tag{9}$$

with

$$G_{HG} = \frac{A_0\beta}{1+\beta+A_0}$$

and

$$\omega_c = \omega_0 \cdot \frac{1+\beta+A_0}{1+\beta}$$

Figure 3 show the Bode plots for the HG op-amp in different gain configurations potentially used by the Super-FGD. The DAC gain value chosen during operation is 44. The low-pass filter behavior is noticeable, with the cutoff frequency changing only slightly for different gains.

2.2 Fast shaper

Now let's look at the Fast shaper. First of all the open loop gain and cutoff pulsation:

$$\omega_0(FSH) = 2\pi \cdot 1.2\ MHz \tag{10}$$

$$A_0(FSH) = 60\ dB = 1000 \tag{11}$$

The closed loop circuit is such as in fig. 4.
We define the time constant of the loop

$$\tau = R_F C_F = R_{in} C_{in} = 2.5\ ns \tag{12}$$

The impedances, with reference to figure 1 are

$$Z_2 = R_F \parallel C_F = \frac{R_F}{1+\tau s}$$

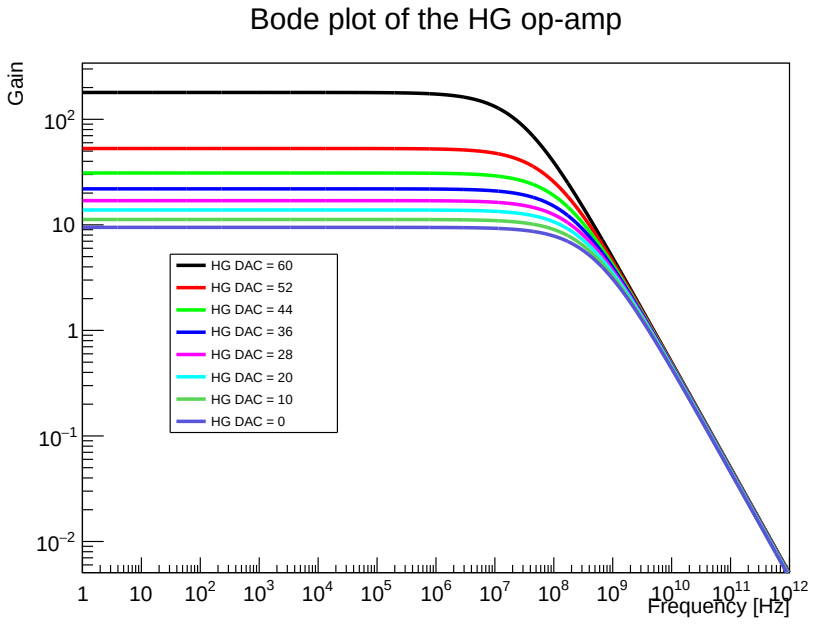


Figure 3: Bode plots of the HG op-amp for different HG DAC values.

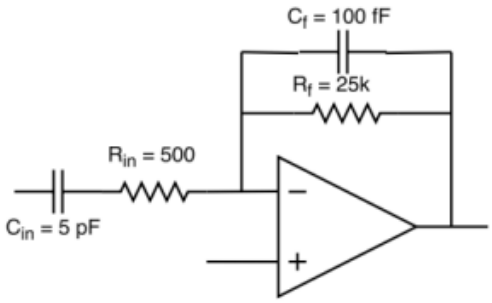


Figure 4: Fast shaper closed loop circuit

We also define the resistance ratio, that determines the gain:

$$\beta_{fast} = R_F/R_{in} = \frac{25k\Omega}{500\Omega} = 50 \tag{13}$$

Plugging everything in equation 3 and working outthe algebra leads to the following transfer function:

$$FSH: \quad TF(s) = \frac{-A_0\beta_{fast}\tau s}{A_0(\tau s+1)^2 + [1+(\beta_{fast}+2)\tau s + \tau^2 s^2](1+\frac{s}{\omega_0})} \tag{14}$$

Let's write down the polynomial in the denominator explicitly

$$TF(s) = \frac{-A_0\beta_{fast}\tau s}{(A_0+1) + [2\tau(A_0+1) + \beta_{fast}\tau + \frac{1}{\omega_0}]s + [t^2(A_0+1) + \frac{\beta_{fast}\tau}{\omega_0} + \frac{2\tau}{\omega_0}]s^2 + \frac{\tau^2}{\omega_0}s^3} \tag{15}$$

Letting $A_0 + 1 \simeq A_0$:

$$TF(s) = \frac{-\beta_{fast}\tau s}{1 + \left[2\tau + \frac{\beta_{fast}\tau}{A_0} + \frac{1}{A_0\omega_0}\right]s + \left[\tau^2 + \frac{\beta_{fast}\tau}{A_0\omega_0} + \frac{2\tau}{A_0\omega_0}\right]s^2 + \frac{\tau^2}{A_0\omega_0}s^3} \tag{16}$$

If we have the computational power to evaluate the inverse Laplace transform numerically, we can use this function as transfer function. In order to have a simpler analytical inverse Laplace transform, and to get an immediate idea of the effect of this circuit, we can write an approximated form. Since the denominator is a second order polynomial, we can decompose into the product of a second order polynomial with two complex roots and a first order polynomial with one real root. This is equivalent to a second order dominant band-pass filter (two complex poles), followed by a first order low-pass filter with larger frequency (one real pole).

$$TF(s) \simeq \frac{-\beta_{fast}\tau s}{(1+2\tau s+4\tau^2 s^2)\left(1+\frac{s}{4A_0\omega_0}\right)} \tag{17}$$

Figure 5 shows the the Bode plot of the actual Fast shaper transfer function (eq. 16) overlaid on the approximated transfer function (eq. 17). For both of them, the gain rises by 20 dB/decade up to a first dominant pole around 20 MHz, then decreases by 20 dB/decade until the second pole, around 30 GHz, then falls by 40 dB/decade. This tells us that the fast shaper can be fairly approximated as a pass-band filter with center frequency $f_p^{FSH} = \frac{1}{2\pi 2\tau} \simeq 20\ MHz$ followed by a low-pass filter with cutoff frequency $f_{cutoff}^{FSH} = \frac{4A_0\omega_0}{2\pi} \simeq 30\ GHz$.

2.3 Timing: global transfer function

The "global" transfer function is determined by the product of the HG and FSH transfer functions. That is the one determining the effective shaping/peaking time of the CITIROC shaped pulse that goes in the discriminator and is used for time stamping. The CITIROC datasheet reports 15 ns as a (rough) estimate of the peaking time, hence a line at 67 MHz is drawn on the plots. Plotting these transfer functions in fig. 6. We can also have a look at how the transfer function changes when we change the (HG) gain, in figure 7 and 8.

3 Finding the time domain response

We can find the inverse Laplace transform numerically, to obtain the response function to a delta pulse and a step function. The peaking time is defined as the time of the first peak of the shaped pulse upon reaction to an ideal step function. This is shown in figure 9 for the case in which HG DAC=44.

It is interesting to show how the circuit responds to an arbitrary current pulse input. In particular, simulating the a typical energy deposit in the super-FGD and the subsequent light propagation in the optic fiber, it is possible to simulate the photon arrival times on the MPPC, which to first approximation

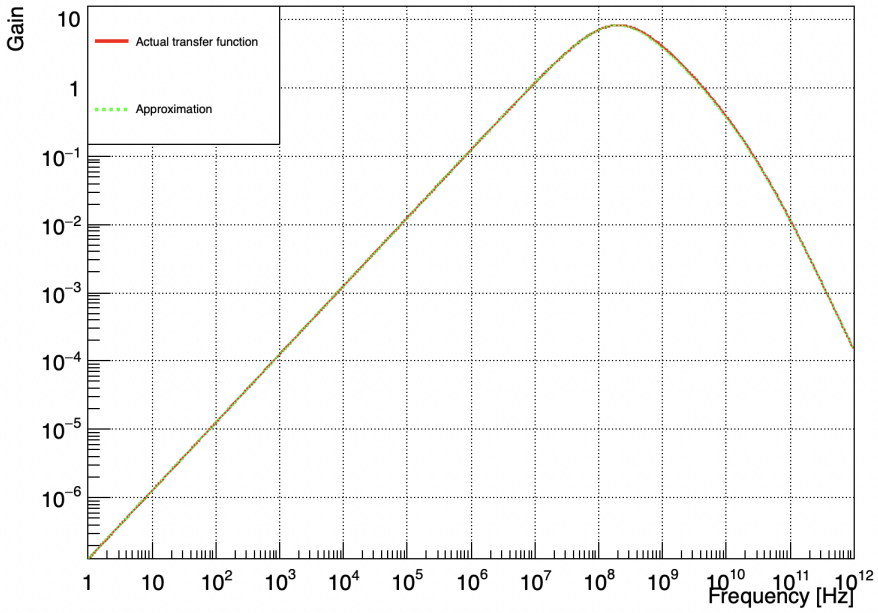


Figure 5: Gain Bode plots for the fast shaper and it's approximation.

is proportional to the current generated at the input of the CITIROC. The shaped pulse generated upon an arbitrary current signal is calculated by numerically convoluting the current waveform and the inverse Laplace transform of the transfer function. Hence, a simple optical + electronics simulation[cite] allows us to do so, and observe the corresponding output of the fast shaper. This is fundamental for the understanding of the time-of-arrival (ToA) and time-over-threshold (ToT) measurements of the CITIROC. Figure 10 shows the current input and the fast-shaped output signal for light generated at different distances from the MPPC. It takes into account the light spreading in the fiber and reflection of the light on the far end of the optic fiber.
It must be noted that this treatment is an approximation. We are neglecting the effect of the MPPC circuit, by saying that the photon distribution in tie is proportional to the current at the input of the CITIROC. To have a better description of this, we should include a simulation of the MPPC response (MPPC "CORSI" model).

4 TODO

- Slow shaper model
- Discriminator model
- MPPC response model
- Write object oriented code (separated models of different components: optics - MPPC - CITIROC)

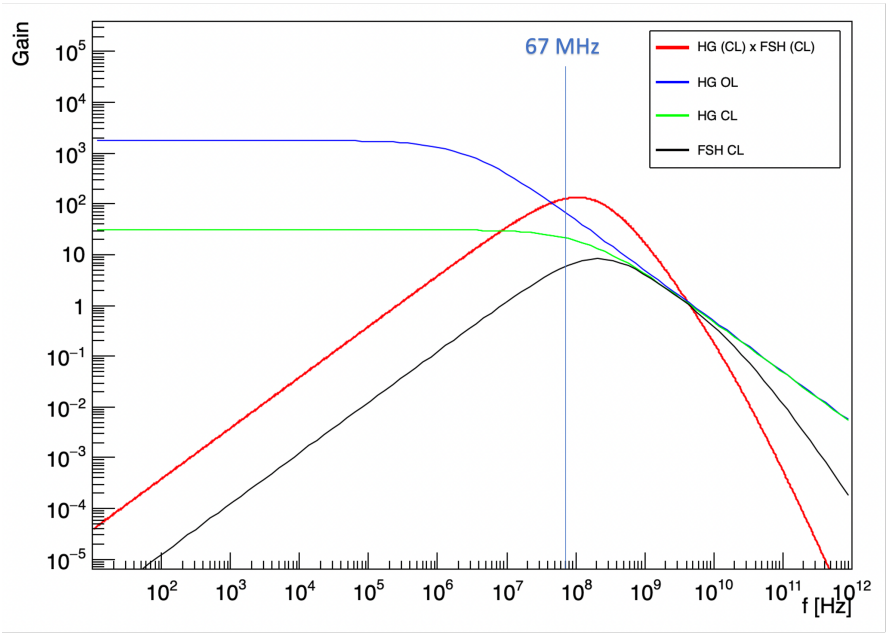


Figure 6: Transfer functions in frequency domain (Bode magnitude plot). In red is the global transfer function for HG·FSH.

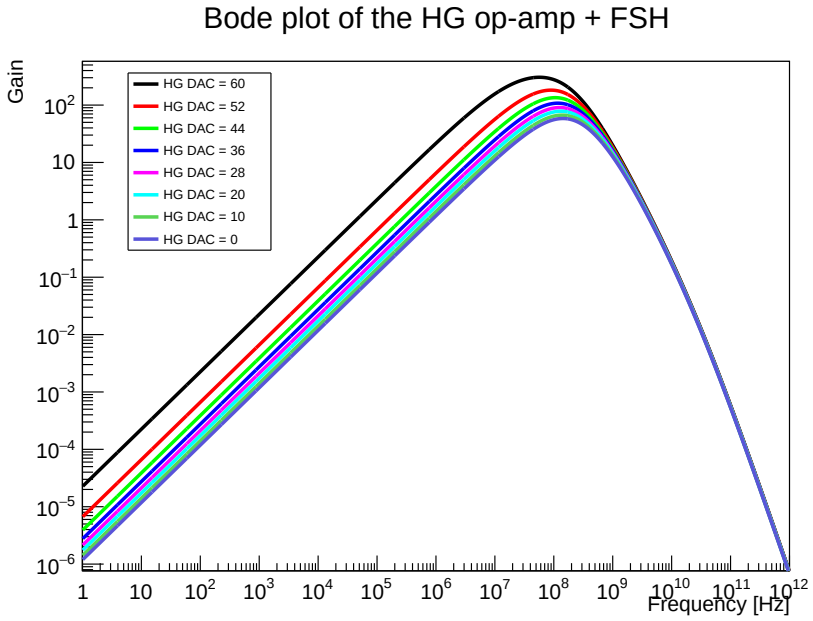


Figure 7: Bode plot of the full time-stamping circuit, comprising HG preamp and fast shaper, for different gain settings.

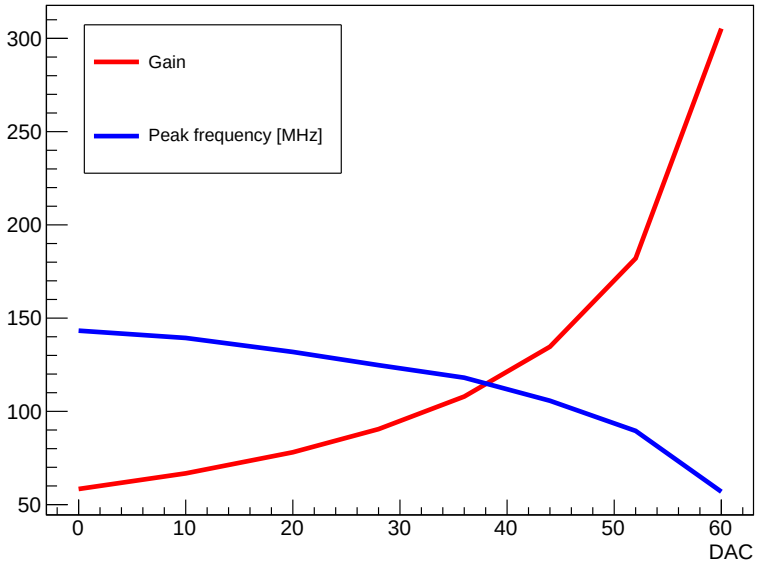


Figure 8: Gain and peak frequency of the time-stamping circuit as a function of teh DAC HG gain setting.

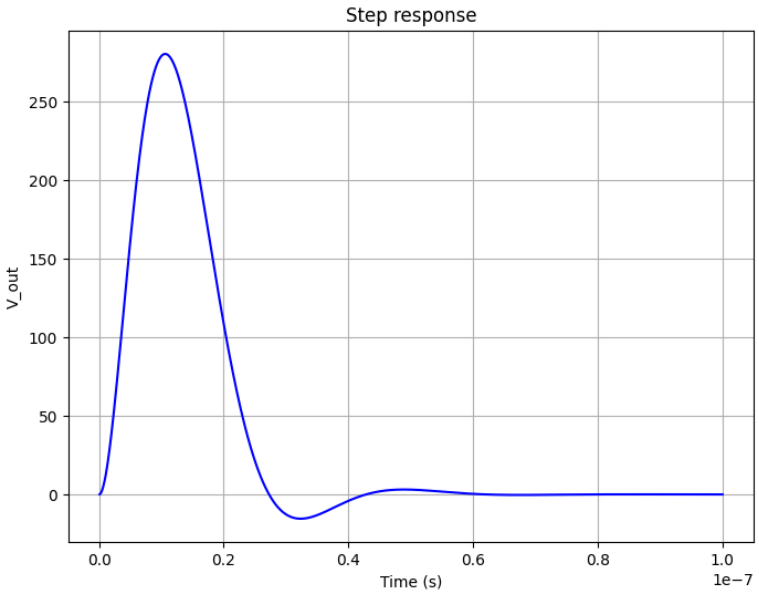


Figure 9: Step function response of the fast shaper. Peaking time: 10.6 ns

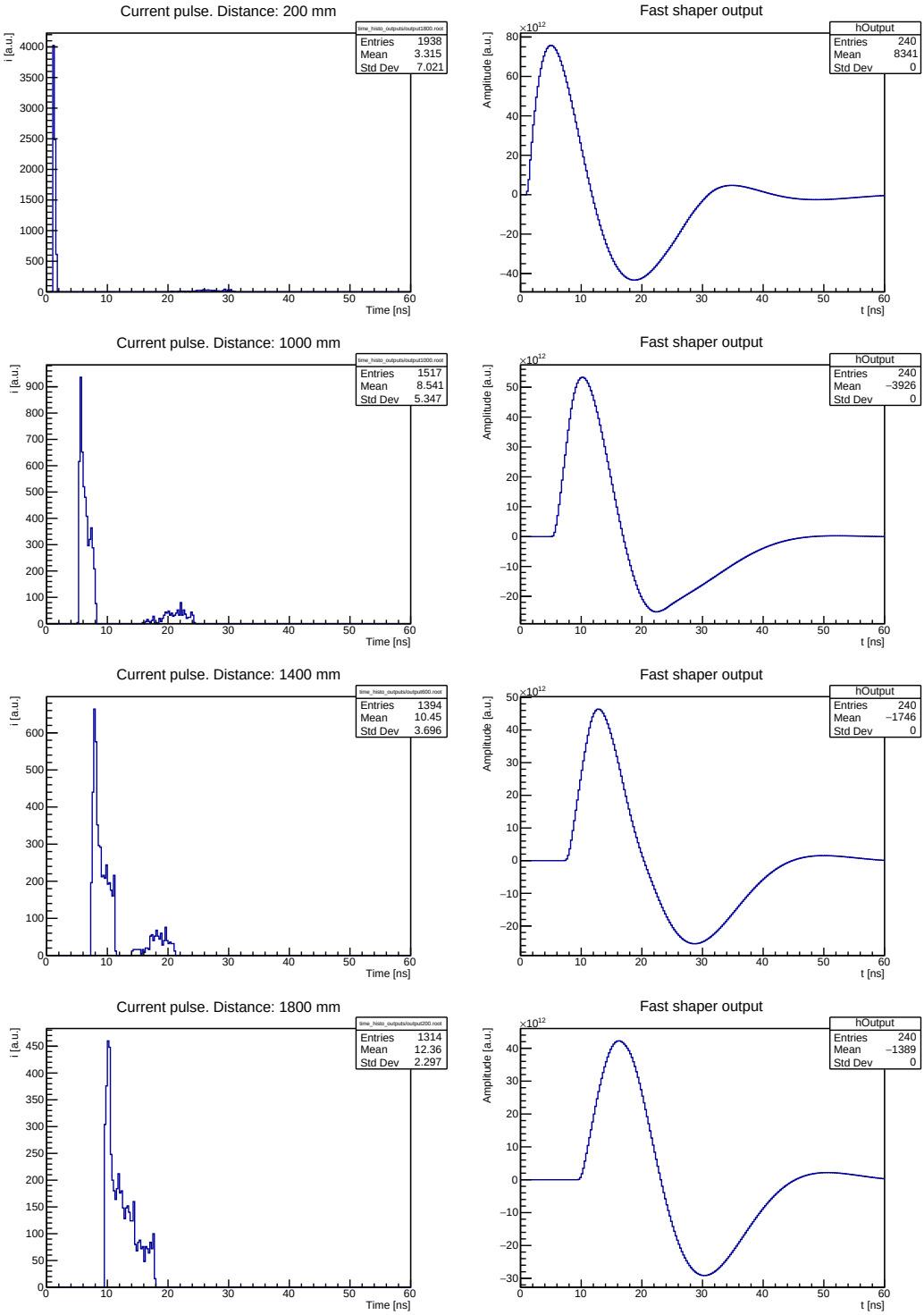


Figure 10: Current pulse and shaped output for different distances of the energy deposit from the MPPC.